[CSED233] Data Structure

Introduction

Seung-Hwan Baek

POSTECH

Course Information: CSED233

- Data Structure
- Tuesday and Thursday 9:30 AM – 10:45 AM
- Building 2, Room 102 (제 2 공학관 102호)

About Instructor

- Seung-Hwan Baek (백승환)
- Associate professor @ Computer Science and Engineering
- Affiliated professor @ Graduate School of Artificial Intelligence
- http://www.shbaek.com
- Research Interests
  - Computational Imaging, Computer Graphics, Computer Vision, Optics, Machine Learning, Robotics
- Office: Rm. 313, PIAI
- E-mail: shwbaek@postech.ac.kr

Teaching Assistants

Hoon-Gyu Chung
(정훈규)

Yonghee Oh
(오용희)

Juhyung Choi
(최주형)

Yujin Jeon
(전유진)

What is Data Structure?

Why Are We Taking Data Structure?

- Programs solve problems by processing information

Program = Data structure + Algorithm

- Data structure is about how information is organized
- Algorithm is about how programs carry out tasks

Data Structure in Modern Era

- There are many open-sourced resources and AI coding agents

GitHub

- The problem you are trying to solve is not exactly same as the ones solved by other resources
- You solutions should be adjusted
- You need to define/understand/modify/implement data structure

AI for Data Structure

- You are encouraged to use any AI models
- POSTECH offers free use of AI models at https://genai.postech.ac.kr/
  - GPT 5.2, Gemini Pro, Sonnet, Deep Research, More to come…
- You are allowed to use AI models to
  - Understand data structure concepts,
  - Learn how to code data structure,
  - Prepare your homeworks
- You are not allowed to use AI models during exam

Example

ChatGPT 4o >

I want to learn data structure. What is stack?

Memory updated

A **stack** is a linear data structure that follows the **Last In, First Out (LIFO)** principle. This means that the last element added to the stack is the first one to be removed.

Key Operations:

1. **Push(x)** – Adds element x to the top of the stack.
2. **Pop()** – Removes and returns the top element of the stack.
3. **Peek()** – Returns the top element without removing it.
4. **isEmpty()** – Checks if the stack is empty.

Example:

Imagine a stack of plates:

- You add (push) a plate on top.
- You remove (pop) the top plate first.

Implementation:

- **Using an array/list** (fixed or dynamic size)
- **Using a linked list** (dynamic memory allocation)

Applications:

- **Function calls (recursion uses a call stack)**
- **Undo/Redo operations** (in text editors, Photoshop, etc.)
- **Expression evaluation** (e.g., converting infix to postfix)

Message ChatGPT

4o

Example

Would you like a code example in Python?

Yes!

Here's a simple **stack implementation in Python** using a list:

Stack Implementation Using List

python Copy

```python
class Stack:
    def __init__(self):
        self.stack = []

    def push(self, item):
        self.stack.append(item)

    def pop(self):
        if not self.is_empty():
            return self.stack.pop()
        return "Stack is empty"

    def peek(self):
        if not self.is_empty():
            return self.stack[-1]
        return "Stack is empty"

    def is_empty(self):
        return len(self.stack) == 0

    def size(self):
        return len(self.stack)

# Example Usage
s = Stack()
s.push(10)
s.push(20)
s.push(30)
print(s.pop())   # Output: 30
print(s.peek())  # Output: 20
print(s.is_empty())  # Output: False
```

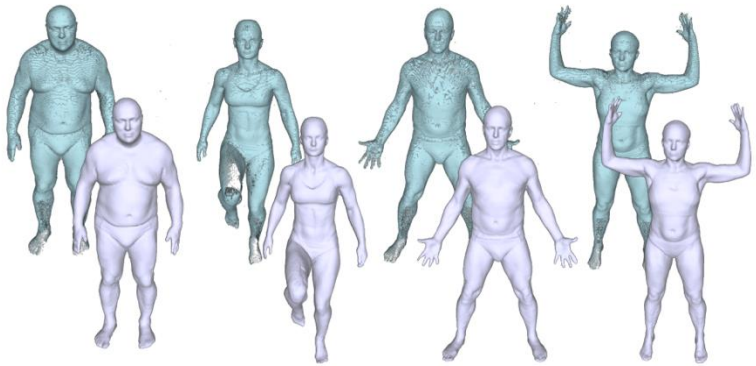
Message ChatGPT

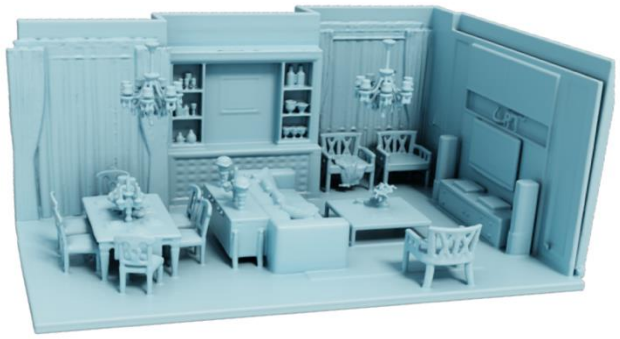
4o

Data Structure for AI

- AI models are specialized to process specific data structure
- It's ok if you don't know each of these models

Data structure	Related AI model
Grid	CNN
Sequence	Transformer/RNN/LSTM/GRU
Set	Deep Sets
Tree	TreeLSTM
Point	PointNet
Graph	GNN

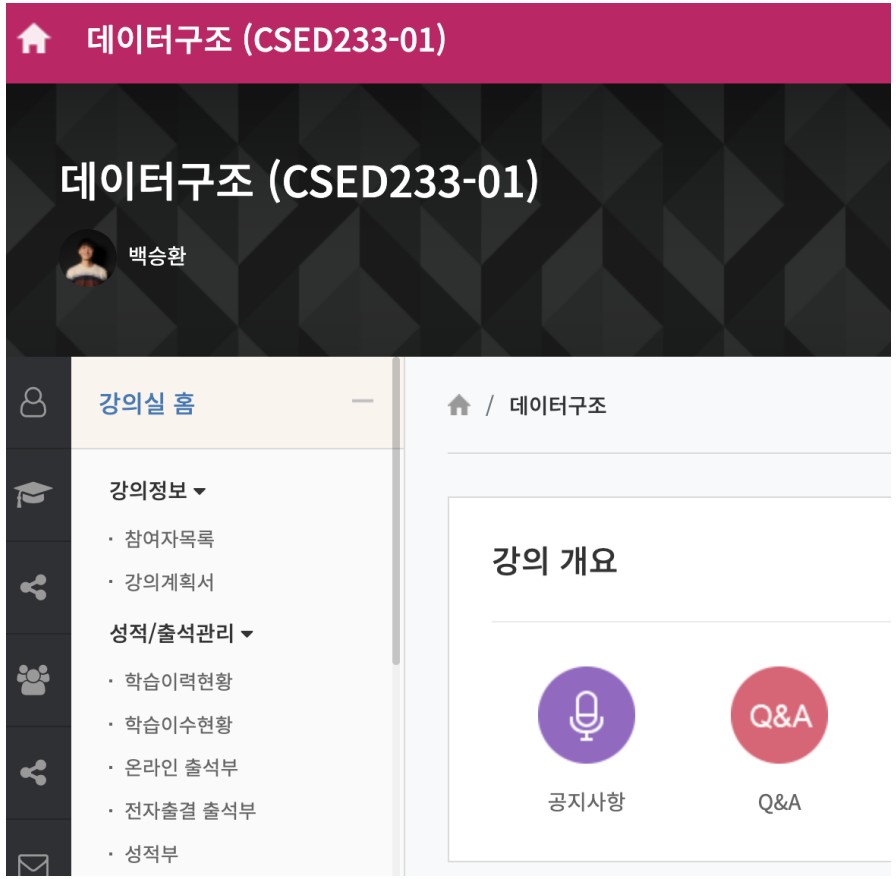




Logistics

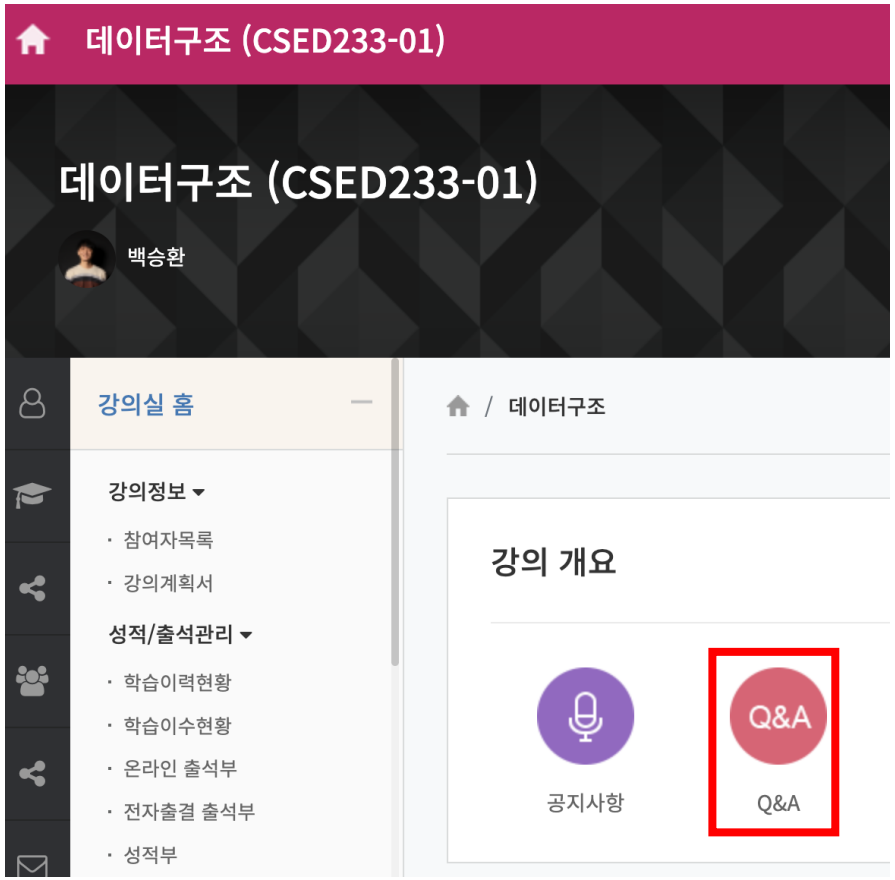
PLMS

- We will use PLMS as the main platform for lecture materials
- Please regularly check our PLMS page



Q/A

- Feel free to ask questions during/after the lecture to me as well
- You are encouraged to ask questions in PLMS for assignments. TAs will answer your questions.



Office Hour

- You can ask questions to TAs on zoom at Mon/Wed 12:15 PM – 13:45 PM
  - https://postech-ac-kr.zoom.us/j/82702734807?pwd=bZhJjNibEESoHbzvI5BXS5v51CcMNw.1
- Zoom session starts from next week
- If you want to have a meeting with me, send me an email

Grading

- Total: 190 pts
  - Four program assignments (PA): 15 pts * 4 =60 pts
  - Midterm: 60 pts
  - Final: 60 pts
  - Course participation: 10 pts
    - Students who ask a question will get additional 2 pts
- If you are taking this course again, you would get the grade up-to B+
- For the students taking this class under S/U evaluation, you need to earn the score that is better than the half of the students

Be Polite and Respect Others

- Professor and TAs
  - We will try to help
  - Please use polite words
- Please respect others
  - Your friend

https://kidshelpline.com.au/teens/issues/all-about-respect

Assignments

- Four programming assignments in C++
- Exploit the concepts taught in the classes
- You can use AI coding models
- Submit though PLMS
- Do not share your code with other students

Exams

- Midterm: 60 pts
- Final: 60 pts
- Some of assignments may be related to the exam questions

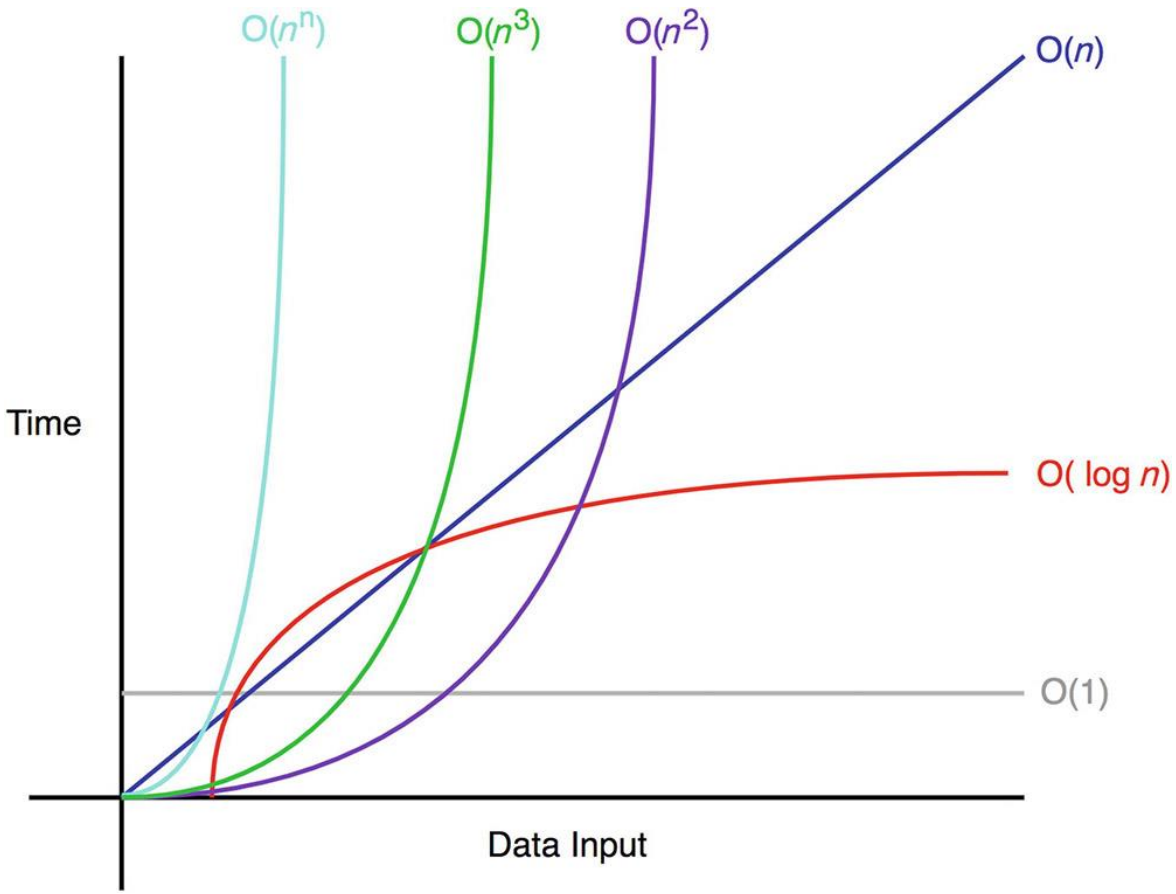
Tentative Fast Forward

C++

- Play with C++ and get ready to do assignments

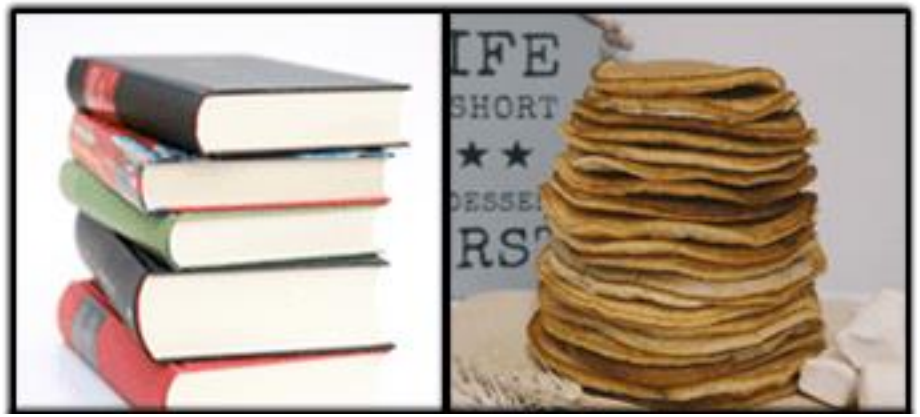
Algorithm Analysis

- Which algorithm is better regardless of the computer spec. you are using?



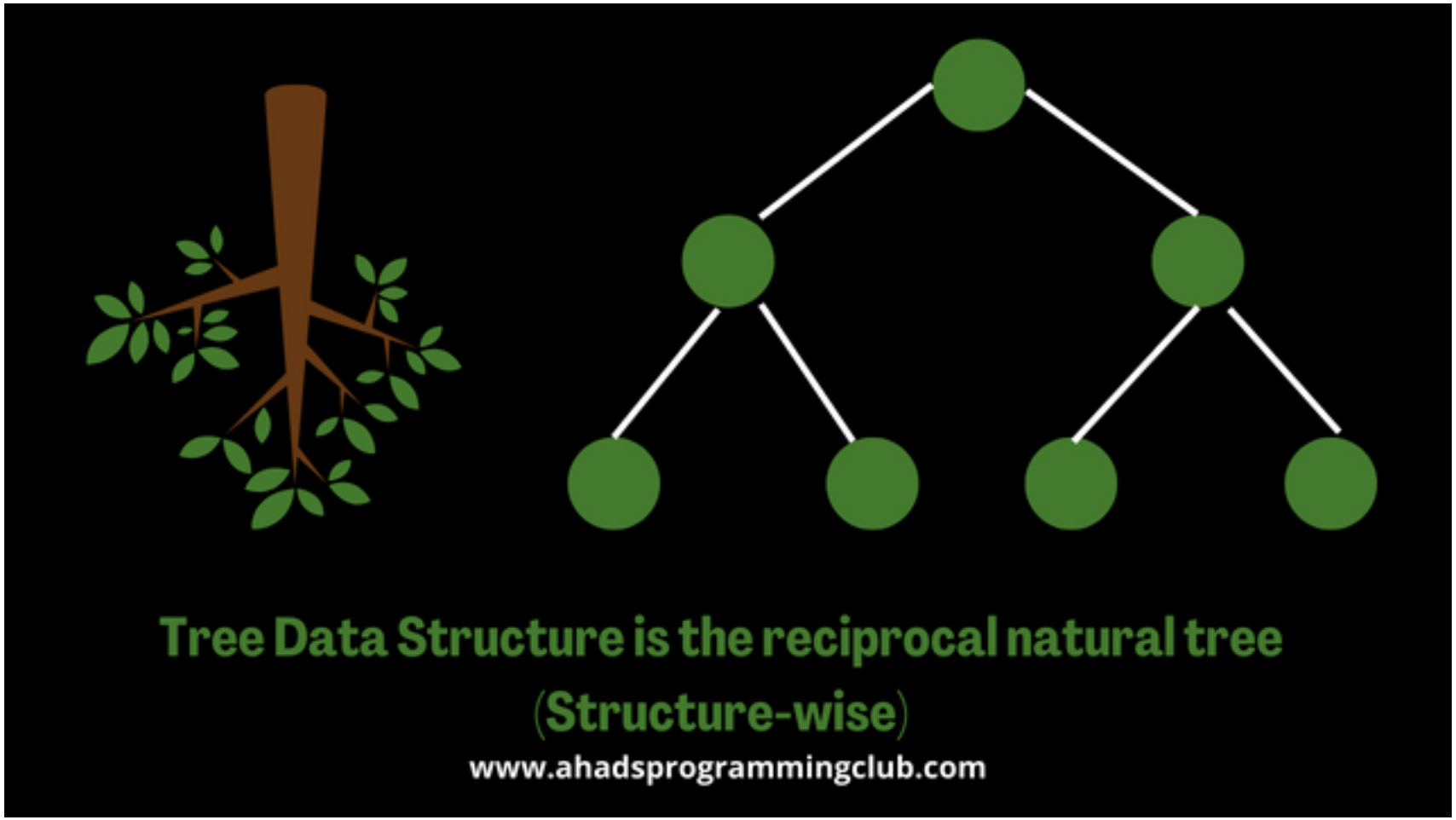
List, Stack, Queue

- List, stack, queue “something” as the names suggest ☺



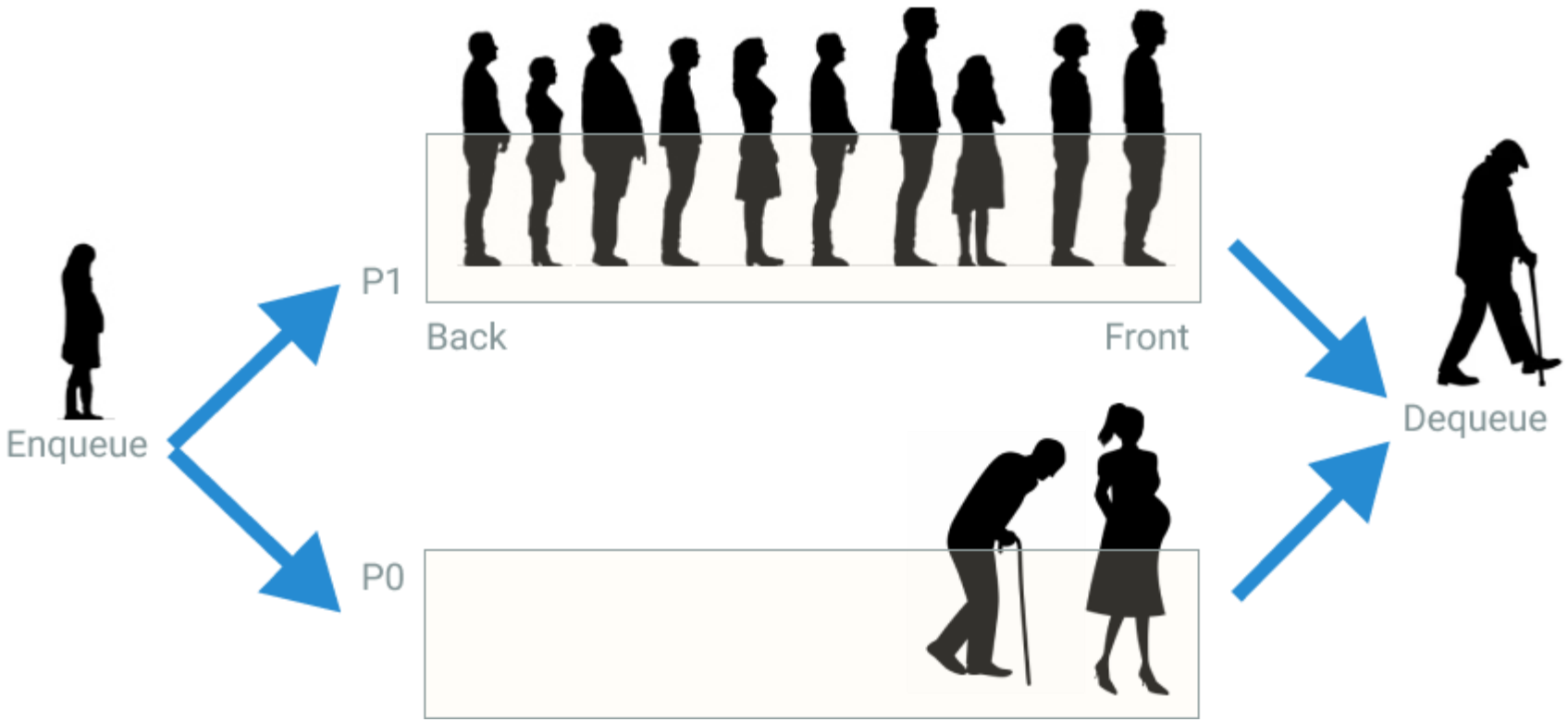
Tree

- Tree data structure looks like a (upside-down) tree
  - Family tree, academic tree, …



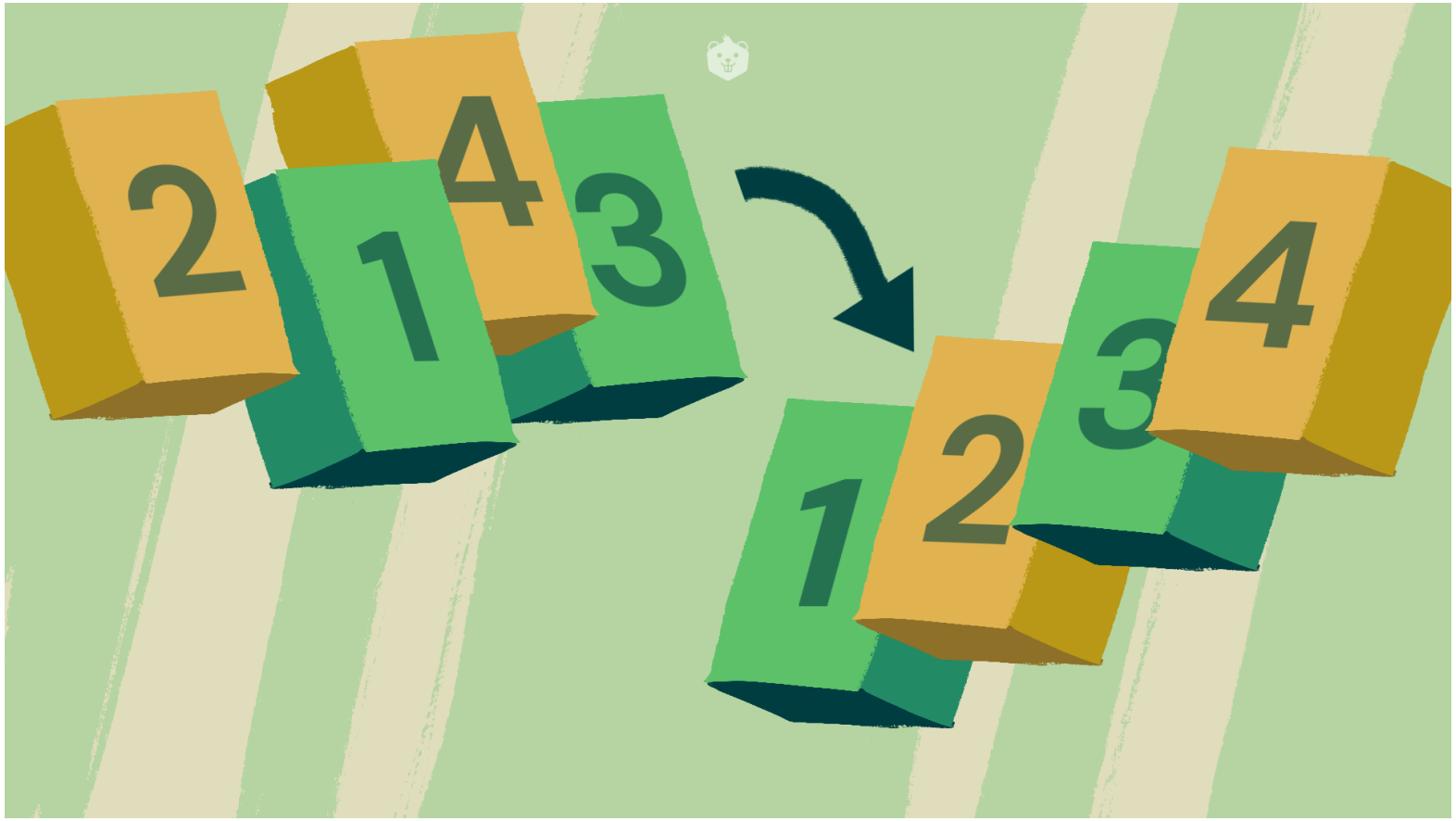
Priority Queue, Heap

- Priority does matter in our life



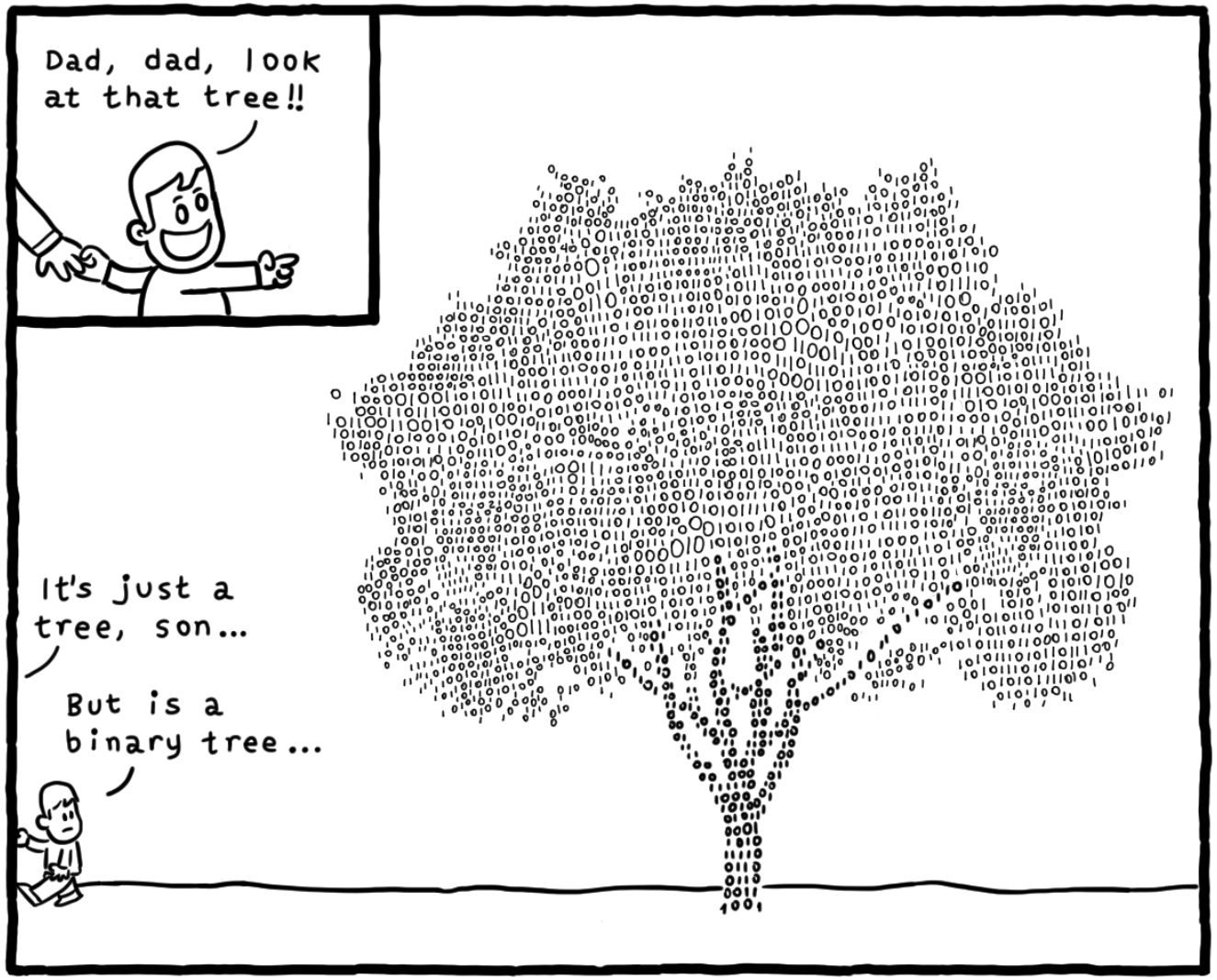
Sorting

- How to computationally make things ordered?



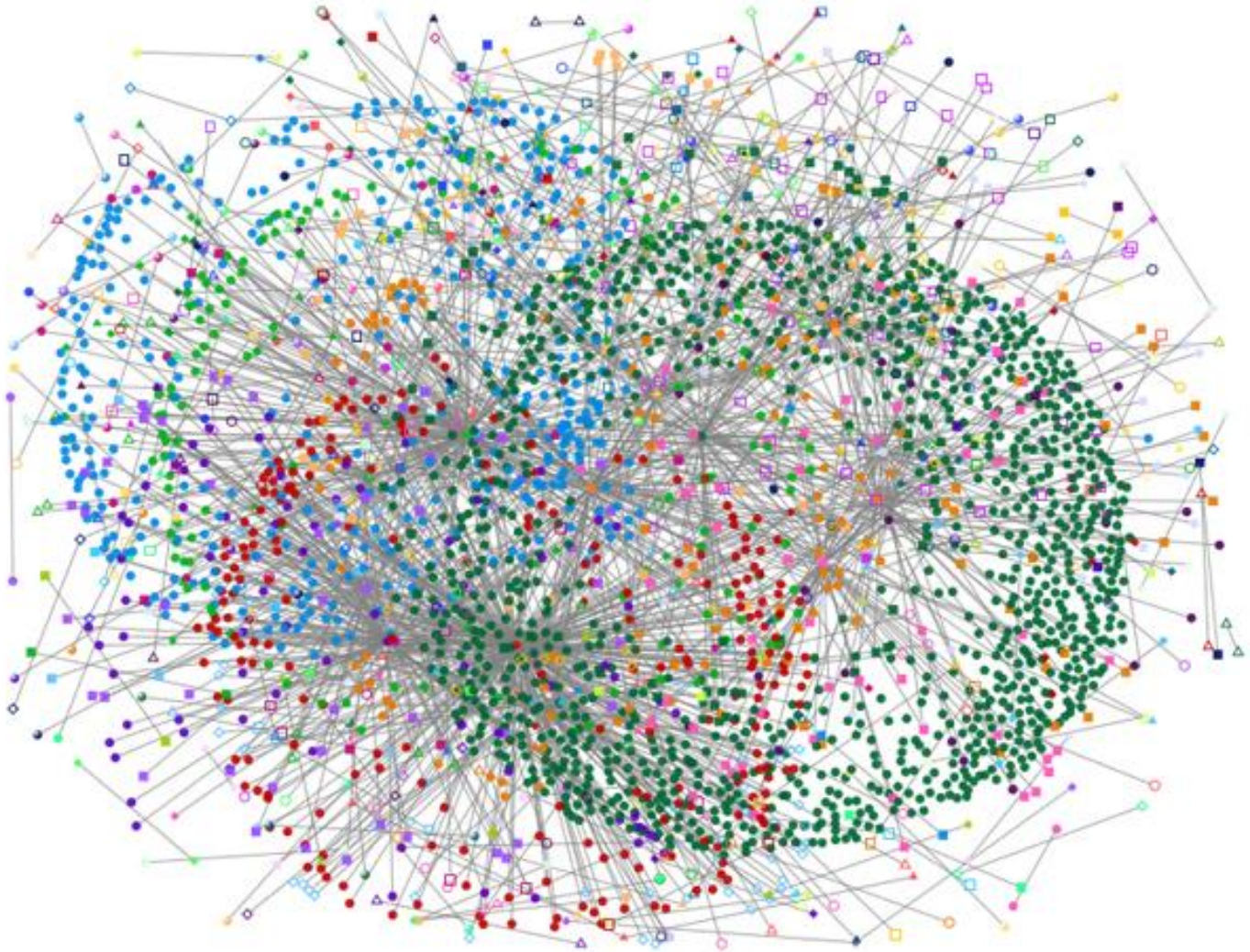
Tree

- There are many special trees that could be useful



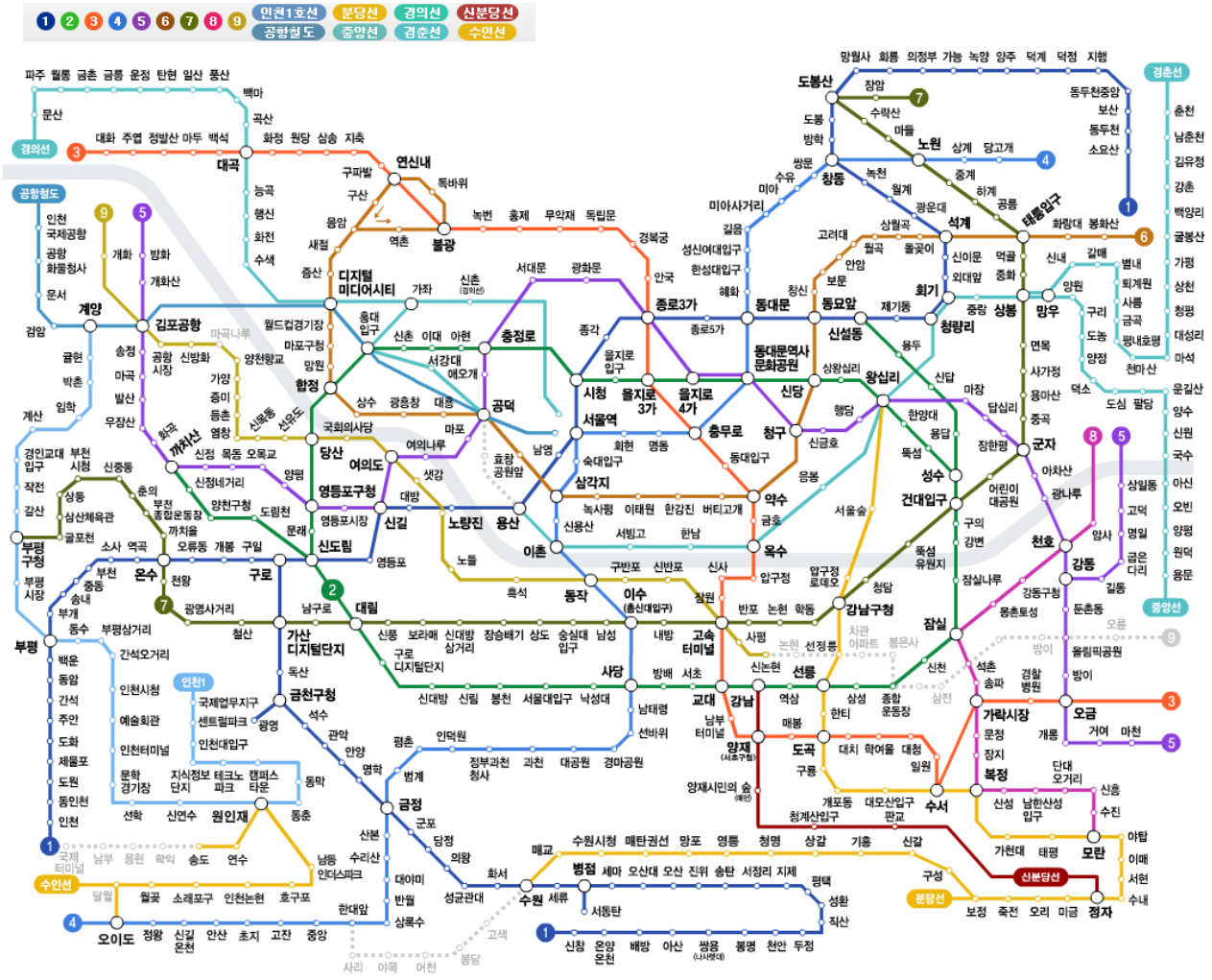
Graph

- Hierarchy may not exist, and anything can be connected



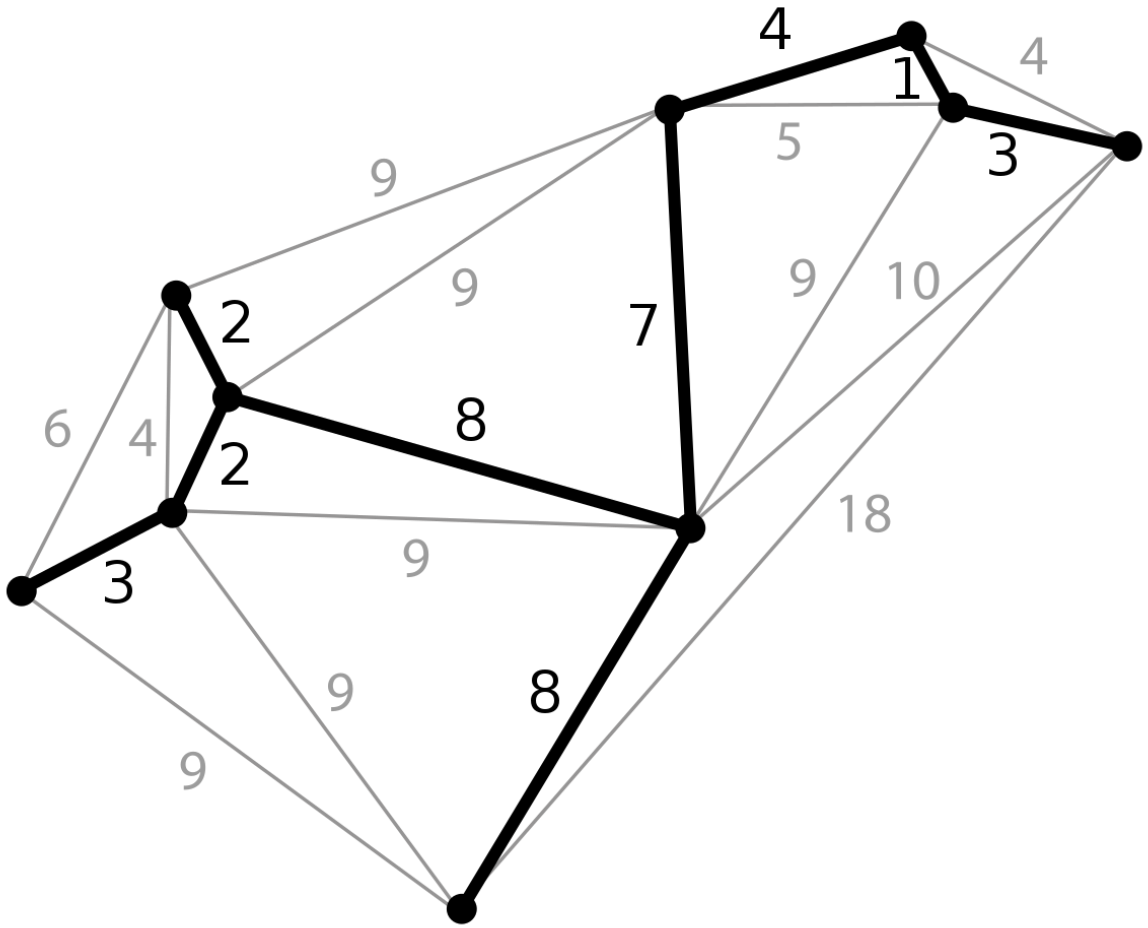
Shortest Path Finding

- You want to go from A to B, in a fastest way


인천1호선
분당선
경의선
신분당선
공항철도
중앙선
경춘선
수인선
연신내
불광
대곡
김포공항
홍대입구
합정
충정로
공덕
시청
을지로3가
을지로4가
서울역
종로3가
동대문
동묘앞
신설동
청량리
회기
석계
태릉입구
노원
창동
도봉산
망우
상봉
왕십리
신당
청구
충무로
약수
옥수
금호
동대입구
삼각지
숙대입구
이촌
용산
노량진
신길
신도림
영등포구청
당산
여의도
까치산
온수
구로
가산디지털단지
대림
남구로
금천구청
석수
관악
안양
범계
평촌
인덕원
금정
군포
당정
의왕
화서
수원
성균관대
세류
병점
서동탄
성환
직산
평택
오이도
정왕
신길온천
안산
초지
고잔
중앙
사리
야목
어천
봉담
고색
한대앞
상록수
대야미
반월
수리산
남동인더스파크
월곶
소래포구
인천논현
호구포
달월
남부
용현
학익
송도
연수
국제업무지구
센트럴파크
광명
인천대입구
인천터미널
문학경기장
지식정보단지
테크노파크
캠퍼스타운
선학
신연수
원인재
동춘
동막
인천시청
예술회관
부평구청
부평시장
부평
계양
귤현
박촌
임학
계산
작전
갈산
굴포천
삼산체육관
상동
송내
부개
동수
부평삼거리
간석오거리
백운
동암
간석
주안
도화
제물포
도원
동인천
인천
인천1
국제터미널
신분당선
분당선
중앙선
경춘선
경의선
공항철도
수인선
춘천
남춘천
김유정
강촌
백양리
굴봉산
가평
상천
청평
대성리
마석
운길산
양수
신원
국수
아신
오빈
양평
원덕
용문
덕소
도심
팔당
양정
도농
구리
양원
별내
퇴계원
사릉
금곡
평내호평
천마산
마석
광릉수목원
퇴계원
별내
갈매
신내
화랑대
봉화산
태릉입구
석계
돌곶이
광운대
월계
하계
공릉
중계
상계
당고개
노원
수락산
도봉
방학
쌍문
수유
미아
미아사거리
길음
성신여대입구
한성대입구
혜화
창신
보문
안암
고려대
월곡
상월곡
신이문
외대앞
먹골
중화
제기동
용두
신답
마장
답십리
장한평
군자
아차산
광나루
어린이대공원
구의
강변
건대입구
성수
뚝섬
서울숲
응봉
금호
옥수
압구정
압구정로데오
신사
잠원
반포
논현
학동
강남구청
청담
삼성
종합운동장
한티
역삼
매봉
도곡
대치
학여울
대청
일원
구룡
개포동
대모산입구
수서
청계산입구
판교
양재
양재시민의숲
남부터미널
교대
강남
서초
방배
사당
내방
이수
총신대입구
남태령
선바위
신대방
신림
봉천
서울대입구
낙성대
인덕원
과천
대공원
경마공원
정부과천청사
범계
평촌
안양
명학
관악
석수
광명
독산
금천구청
가산디지털단지
구로디지털단지
신대방삼거리
장승배기
상도
숭실대입구
남성
숭실대입구
신풍
보라매
신대방
신림
봉천
서울대입구
낙성대
사당
방배
서초
교대
강남
역삼
삼성
종합운동장
한티
매봉
도곡
구룡
대치
학여울
대청
일원
개포동
대모산입구
수서
청계산입구
판교
양재
양재시민의숲
남부터미널
신논현
논현
선정릉
삼전
석촌
송파
경찰병원
방이
올림픽공원
오금
거여
마천
가락시장
문정
장지
복정
단대오거리
신흥
수진
산성
남한산성입구
가천대
태평
모란
야탑
이매
서현
수내
정자
보정
죽전
오리
미금
구성
판교
삼평동
봇들
백현
분당
서현
이매
선릉
역삼
강남
교대
서초
방배
사당
내방
이수
총신대입구
남태령
선바위

Minimum Spanning Tree

- How to connect all the places with minimum cost?



String

- Trie, suffix tree, pattern matching